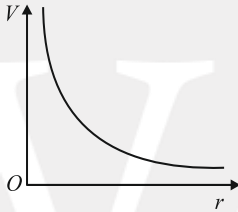
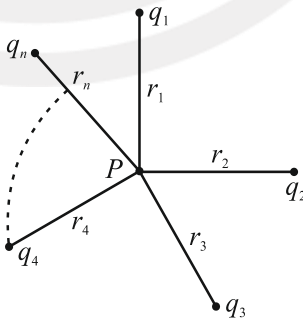
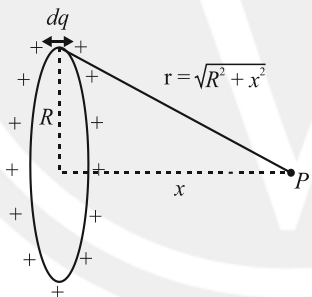


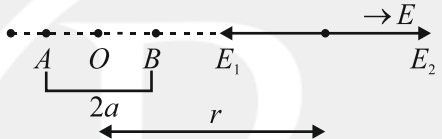
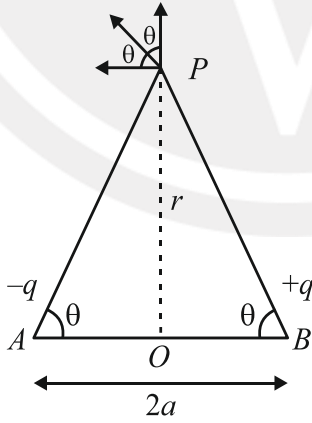
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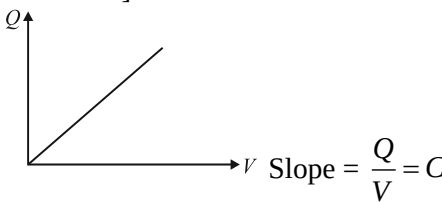
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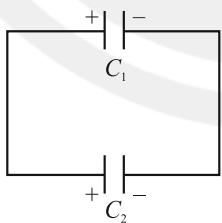
Electrostatic Potential and Capacitance

Sr. No.	Concept	Formulas	Other information
1.	Electric potential	Electric potential (V) = $\frac{\text{Work done (W)}}{\text{Charge}(q_0)}$ 1 volt = $\frac{1}{300}$ stat volt Dimensional formula is $[ML^2T^{-3}A^{-1}]$.	
2.	Electrostatic potential due to a point charge	For the definition of electric potential, $\therefore V = \frac{W}{q_0} = \frac{q}{4\pi\epsilon_0 r}$ for $q > 0, v > 0, q < 0, v < 0$ Potential with distance, i.e., $V \propto \frac{1}{r}$. 	V = electric potential W = work done q = electric charge
3.	Electrostatic potential due to a system of charges	$V = V_1 + V_2 + V_3 + \dots + V_n$ $V = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r_{1P}} + \frac{q_2}{r_{2P}} + \frac{q_3}{r_{3P}} + \dots + \frac{q_n}{r_{nP}} \right)$ $V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_{iP}}$ 	V = potential due to charges $q_1, q_2, q_3, \dots, q_n$ at distances $r_1, r_2, r_3, \dots, r_n$ respectively from the point P
4.	Electrostatic potential energy (U)	Electric potential energy of two-point charges, $U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r_{12}}$	q_1, q_2 = two-point charges separated by distance r_{12}

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5.	Electrostatic potential energy due to dipole (U)	$U = -pE \cos \theta$ or $U = -\vec{p} \cdot \vec{E}$. Case-(i): When $\vec{E} \parallel \vec{p} \Rightarrow \theta = 0^\circ$ $\Rightarrow U = -pE = U_{\min}$ $(\because \text{Here } F = 0 \Rightarrow \text{stable equilibrium})$ Case-(ii): When $\vec{E} \perp \vec{p} \Rightarrow \theta = 90^\circ$ $\Rightarrow U = 0$ (reference position) (Here F = maximum)	E = conservative field. p = electric dipole moment. U = potential energy
6.	Finding E when V is given:	Gradient $\rightarrow$ Rate of change with distance $\Rightarrow E = \frac{-dV}{dr}$ or $\vec{E} = \frac{-dV}{dr} \hat{r}$. In cartesian system $\vec{E} = -\nabla V$ and $\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$ $\vec{E} = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$ Electric field in the space is $\vec{E} = E_x \hat{i} + E_y \hat{j} + E_z \hat{k}$	E = electric field intensity. V = electric potential
7.	Electrostatic potential due to a uniformly charged ring	Potential due to the ring on its axis: $V = \frac{Q}{4\pi\epsilon_0(R^2 + x^2)^{\frac{1}{2}}} \Rightarrow V = \frac{kQ}{\sqrt{R^2 + x^2}}$  Putting $x = 0$ and potential at the centre of ring, $V_{\text{center}} = \frac{kQ}{R}$	k = dielectric constant. Q = total charge on the ring R = radius of the ring. x = distance of the calculated point.
8.	Electrostatic potential due to a charged conducting sphere	[a] At an extreme point ($r > R$), $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ [b] At the surface of a shell ($r = R$), $V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$ [c] At an internal point ($r < R$), $V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$	V = electrostatic potential due to charged conducting sphere. q = total charge on the sphere. R = radius of the conducting sphere r = internal point distance from the centre.

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9.	Electric potential due to uniformly charged non-conducting sphere	[a] At an extreme point, ($r > R$) $\text{Electric potential } V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ [b] On the surface, ($r = R$) $\text{Electric potential } V = \frac{1}{4\pi\epsilon_0} \frac{q}{R}$ [c] Inside the sphere, ($r < R$) $\text{Electric potential } V = \frac{1}{4\pi\epsilon_0} \frac{q(3R^2 - r^2)}{2R^3}$	V = electrostatic potential of charged non-conducting sphere. q = total charge on the sphere. R = radius of the non-conducting sphere. r = internal point distance from the centre
10.	Electric potential due to electric dipole	Electric potential $V_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 - a^2)}$ If $r \gg 2a$, then $V_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$  At an Equatorial Point of a dipole: Electric field intensity $E_{\text{equa}} = \frac{1}{4\pi\epsilon_0} \frac{p}{(r^2 + a^2)^{\frac{3}{2}}}$ If $r \gg 2a$, then $E_{\text{equa}} = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3}$ Electric potential $V_{\text{equa}} = 0$ 	$2a$ = distance between the two electric charge. r = distance from the midpoint of the dipole to external point. p = electric dipole moment.
11.	Work done in rotating a dipole in a uniform electric field (W)	Total work done by external forces in rotating a dipole from $\theta = \theta_1$ to $\theta = \theta_2$ will be given by $W = \int_{\theta_1}^{\theta_2} pE \sin \theta d\theta$ $W_{\text{external forces}} = pE(\cos \theta_1 - \cos \theta_2)$	W = work done p = electric dipole moment. E = electric field

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12.	Electrical capacitance (C)	$C = \frac{Q}{V} \Rightarrow \text{constant [electrical capacitance of a conductor]}$ 	$Q = \text{electric charge}$ $V = \text{electric potential}$
13.	Energy stored in a capacitor	$\text{P.E.} = W_{\text{ext.}}$ $U = \frac{1}{2} CV^2 = \frac{1}{2} QV = \frac{Q^2}{2C}$ Energy stored per unit volume or the energy density of the electric field of a capacitor, $u = \frac{1}{2} \epsilon_0 E^2$	$u = \text{energy density of the capacitor.}$ $E = \text{electric field}$ $Q = \text{total charge of the capacitor}$ $V = \text{potential difference between the plates.}$ $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
14.	Spherical capacitance	Potential $V = \frac{kQ}{R}$, Hence by formula: $Q = CV$ $\Rightarrow Q = \frac{CKQ}{R} \Rightarrow C = 4\pi \epsilon_0 R$ When outer sphere is earthed: Potential of inner sphere $V_1 = \frac{Q}{4\pi \epsilon_0 a} + \frac{-Q}{4\pi \epsilon_0 b}$ $\Rightarrow \frac{Q}{4\pi \epsilon_0} \left(\frac{b-a}{ab} \right)$ The capacitance of the spherical capacitor is $C = 4\pi \epsilon_0 \left(\frac{ab}{b-a} \right)$	$R = \text{radius of spherical conductor.}$ $K = \text{dielectric constant.}$ $b = \text{radius of the outer sphere}$ $a = \text{radius of the inner sphere}$ $Q = \text{charge on the spherical conductor.}$
15.	Cylindrical capacitor	Potential difference between plates $V = \int_a^b \frac{Q}{2\pi \epsilon_0 r \ell} dr = \frac{Q}{2\pi \epsilon_0 \ell} \ln \left(\frac{b}{a} \right)$	$a, b = \text{radius of cylinders.}$ $\ell = \text{length of the cylinders.}$
16.	Parallel plate capacitor	Capacitance of a parallel plate capacitor, $C = \frac{\epsilon_0 A}{d}$ P.D. between the two plates of a capacitor having charges q_1 and q_2 , $V = \frac{q_1 - q_2}{2C}$ Electric field in dielectric medium, $E_m = \frac{\sigma}{\epsilon} = \frac{q}{\epsilon_0 \epsilon_r A}$	$A = \text{overlapping area}$ $d = \text{distance between plates}$ $\epsilon = \text{permeability of medium between plates}$

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17.	Sharing of charges (Common potential)	Common potential after connection $V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$ Charges after connection $Q_1 = C_1 V = C_1 \left(\frac{Q_1 + Q_2}{C_1 + C_2} \right) = \left(\frac{C_1}{C_1 + C_2} \right) Q$ (Q = Total charge on the system) $Q_2 = C_2 V = C_2 \left(\frac{Q_1 + Q_2}{C_1 + C_2} \right) = \left(\frac{C_2}{C_1 + C_2} \right) Q$ Ratio of the charges after redistribution $\frac{Q_1'}{Q_2'} = \frac{C_1}{C_2} = \frac{R_1}{R_2} \text{ (in case of spherical conductors)}$	C_1, C_2 = capacitance of two sphere. Q_1' and Q_2' = amounts of charges after the conductors are connected respectively. V = common potential.
18.	Combination of capacitors-series and parallel	Series combination- $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \text{ or in general } \frac{1}{C_{eq}} = \sum_{n=1}^n \frac{1}{C_n}$ Energy stored in the combination $U_{\text{combination}} = \frac{Q^2}{2C_1} + \frac{Q^2}{2C_2} + \frac{Q^2}{2C_3}$ Parallel combination- $C_{eq} = C_1 + C_2 + C_3$ In general, $C_{eq} = \sum_{n=1}^n C_n$	Total charge $\Rightarrow Q = Q_1 + Q_2 + Q_3 \dots\dots$
19.	For parallel plate capacitor (common potential)	$V_{cm} = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$ $\text{Heat loss} = \frac{1}{2} \frac{C_1 C_2}{(C_1 + C_2)} (V_1 - V_2)^2$ 	C_1, C_2 = capacitance of the two capacitors. V_1, V_2 = potentials of the two capacitors.
20.	When small drops coalesce to form a single drop	Total charge = nq Total capacitance = $n^{1/3}C$ Total potential = $n^{2/3}V$ Surface charge density $n^{1/3}\sigma$ Total potential energy = $n^{2/3}U$	n = number of small drops each of capacitance C charged to potential V with charge q. σ = surface charge density U = potential energy